

InfraWatch Nexus

AI-Powered Real-Time Urban Sanitation & Infrastructure Command Center

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Live: <https://infrawatch-nexus-tnlf.onrender.com>

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1. Abstract & Problem Statement

1.1 Abstract

InfraWatch Nexus is a production-grade, real-time AI command center for urban sanitation and infrastructure management. It connects citizens directly to municipal dispatch operations through streaming event architecture, computer vision AI, and live weather-aware risk scoring. The system uses the Pathway streaming engine for real-time event aggregation, Google Gemini 2.5 Flash for AI-powered waste detection from citizen photos, and live weather data from WeatherAPI.com to dynamically prioritize infrastructure issues across Delhi's 12 municipal wards.

1.2 Problem Statement

Traditional municipal reporting in Indian cities is reactive, fragmented, and blind to real-time conditions:

Problem	Impact
Citizens fill lengthy complaint forms	0% transparency, reports lost in bureaucracy
Garbage trucks follow static schedules	Wasted fuel, higher emissions
Road hazards not mapped dynamically	3,500+ deaths/year (MoRTH data)
No weather integration	Blocked drains = epidemic risk during rain

InfraWatch Nexus replaces all of this with a single AI-powered, weather-aware, real-time command center.

2. System Architecture

InfraWatch Nexus follows a strict separation-of-concerns architecture with four distinct layers. Each layer has a single, well-defined responsibility:

Layer	Does	Does NOT
Citizens' Portal	Accept photo, show live state	Compute anything
Admin Portal	Report road issues, dispatch vans	Compute anything
FastAPI Server	Validate, write events, auth, broadcast	Score, rank, aggregate
Pathway Engine	Aggregate, score, rank, weather join	Serve HTTP, touch frontend
WebSocket	Broadcast atomic state to all clients	Compute or filter

2.1 Architecture Flow

1. Citizen uploads a photo via the Citizens' Portal (SPA)
2. FastAPI sends the image to Gemini 2.5 Flash Vision API for asset ID extraction
3. AI returns the exact MCD dustbin ID (e.g., MCD-W04-001)
4. Citizen confirms the detection, and FastAPI writes a JSON event to the data directory
5. Pathway Streaming Engine watches the data directory and recomputes the dashboard
6. Live weather data from WeatherAPI.com is polled every 10 minutes and used as a risk multiplier
7. Pathway computes risk scores, state transitions, and priority queue ordering
8. The atomic dashboard snapshot is broadcast via WebSocket to all connected portals
9. Admin can dispatch vans, clear issues, and manage the priority queue in real-time

2.2 Key Design Principles

- **Transport-only:** Zero computation in FastAPI —
- **Single brain:** All scoring, aggregation, and state management lives in Pathway
- **Atomic writes:** Dashboard state written via temp-file + `os.replace()` — no partial reads
- **Event-driven:** Single WebSocket channel broadcasts identical state to all clients

3. Core Features

3.1 AI-Powered Citizen Reporting

- Citizens upload a single photo — AI (Gemini 2.5 Flash Vision) extracts the exact MCD dustbin ID
- Zero friction: No forms, no dropdowns. One photo = one verified report
- Manual fallback: If AI fails, citizen gets a ward-filtered dropdown for manual selection
- Geolocation-aware: GPS coordinates are captured and stored with each report

3.2 Pathway Streaming Engine (The Brain)

- Event-time windowing: 2-hour rolling windows for waste reports, 6-hour for road issues
- Dustbin State Machine: Clear > Reported > Escalated > Critical > Cleared
- Weather-aware risk scoring: Live rainfall acts as a multiplier on risk scores
- Atomic JSON output: Dashboard state written via temp-file + `os.replace()`
- 3-second recompute loop ensures dashboard always reflects latest state

3.3 Admin Command Center

- Live Priority Dispatch Queue: Auto-sorted by dynamic risk score (0-100)
- Interactive OSRM-Routed Map: Road hazards rendered as real street-level polylines
- Clear Issues Panel: 1-click resolution of dustbins and road hazards
- Simulate Crisis: Demo button injects severe events for live demonstration
- Predictive Risk Forecasting: ML-powered 3-day risk prediction using weather forecast

3.4 Real-Time WebSocket Sync

- Single WebSocket channel broadcasts identical atomic state to all connected portals
- Auto-reconnect with exponential backoff on disconnection
- Citizens' and Admin maps update simultaneously within milliseconds

3.5 Security & Auth

- Admin endpoints protected by Bearer token authentication (strict 401 on failure)
- In-memory $O(1)$ dedup prevents duplicate reports within 5-minute windows
- Dustbin ID validation via strict regex (`MCD-W\d{2}-\d{3}`) against the MCD registry

4. Technology Stack

Technology	Version	Purpose
Python	3.10+	Backend runtime
FastAPI	>=0.110.0	Async web framework & WebSocket server
Pathway	>=0.14.0	Real-time streaming data engine
Gemini 2.5 Flash	Latest	Computer vision for waste detection
WeatherAPI.com	v1	Live rainfall data integration
Leaflet.js	1.9+	Interactive map rendering
OSRM	Latest	Open-source road routing engine
pdfplumber	-	Government PDF data extraction
GitHub Actions	-	CI/CD pipeline
Docker	Latest	Containerized deployment
Render.com	-	Cloud hosting platform
Uvicorn	>=0.29.0	ASGI server for production

4.1 Why Pathway?

Pathway is a real-time data processing engine that provides event-time windowing, incremental computation, and streaming table semantics. Unlike batch-based systems (e.g., Spark), Pathway reacts to file changes in real-time, making it ideal for our event-driven architecture where every citizen report triggers an immediate dashboard update. The engine watches data directories and automatically triggers recomputation when new events arrive, ensuring sub-second latency from report submission to dashboard update.

4.2 Why Gemini 2.5 Flash?

Google's Gemini 2.5 Flash model provides best-in-class vision capabilities at the free tier (15 requests per minute). The model extracts structured data (MCD asset IDs) from citizen photos with high accuracy, enabling zero-friction reporting without forms or dropdowns. The free tier is sufficient for pilot city deployment, and paid tiers scale linearly.

5. Data Sources & Credibility

All infrastructure data in this project is sourced from official government records.

Data Layer	Source	Type
Dustbin / Dhalao Locations	MCD - RO No. 20/DPI/MCD/2024-25	Official Govt PDF
Weather (Rainfall)	WeatherAPI.com - Live polling 10 min	Real-time API
Citizen Reports	Live user submissions - Gemini Vision	Real-time user data
Road Hazard Reports	Admin-submitted - GPS-tagged	Real-time admin data

5.1 MCD C&D Waste Collection Sites

The 72-point dustbin registry (config/dustbins.py) is built from the official MCD document listing 106 designated C&D (Construction & Demolition) waste collection sites across all Delhi zones. Source: RO No. 20/DPI/MCD/2024-25 published by Municipal Corporation of Delhi (mcdonline.nic.in). Data was extracted programmatically using pdfplumber and geocoded for spatial analysis using verified Delhi GPS coordinates.

Zone	Area	Example Site
Rohini	North Delhi	JE Store, Sector-5 Rohini
Karol Bagh	Central-West	MCD JE Store, East Patel Nagar
Shahdara South	East Delhi	Karkari Mod, Karkardooma
South	South Delhi	JE Store, Hauz Khas Market
Keshav Puram	North-West	JE Store, Pitampura
Central 1	Central	Defence Colony, Srinivaspuri
Civil Lines	North-Central	Qutab Road, Burari
City SP	Old Delhi	Chandni Chowk, Asaf Ali Road
South 1	Far South	Fatehpur Beri, Khanpur
Narela	Far North	MPL Store, Nehru Enclave
Central	Central	Minto Road, Punjabi Bagh
Shahdara North	North-East	Seelampur, Jafrabad

6. Risk Scoring Algorithm

InfraWatch Nexus uses a weighted multi-factor risk scoring system to compute dynamic risk scores (0-100) for each ward. Separate algorithms handle waste risk and road risk.

6.1 Waste Risk Score

Formula: $\text{waste_risk} = \text{SUM}(\text{weight_i} * \text{normalized_factor_i}) * 100$

Factor	Weight	Normalization Threshold
Report Frequency (2hr window)	0.35 (35%)	8+ reports = maximum
Overflow Severity (1-5)	0.30 (30%)	Level 5 = overflowing
Collection Delay (hours)	0.20 (20%)	12+ hours = maximum
Rainfall (mm/hr)	0.15 (15%)	50mm/hr = max (capped)

6.2 Road Risk Score

Formula: $\text{road_risk} = \text{SUM}(\text{weight_i} * \text{normalized_factor_i}) * 100$

Factor	Weight	Normalization Threshold
Report Density (6hr window)	0.60 (60%)	6+ reports = maximum
Issue Severity (1-5)	0.25 (25%)	Severity 5 = critical
Rainfall (mm/hr)	0.15 (15%)	50mm/hr = max (capped)

6.3 State Bands

Score Range	Label	Color
0 - 30	Normal	#16A34A (Green)
31 - 55	Elevated	#D97706 (Amber)
56 - 75	Warning	#EA580C (Orange)
76 - 100	Critical	#DC2626 (Red)

A hysteresis buffer of 10 points prevents oscillation between states at boundary values.

7. Dustbin State Machine

Each dustbin in the MCD registry follows a finite state machine that transitions based on report count, overflow severity, rainfall conditions, and van collection events.

State	Trigger	Visual
Clear	0 reports in rolling window	Green marker
Reported	1+ report(s) received	Yellow marker
Escalated	3+ reports OR overflow ≥ 4	Orange marker
Critical	5+ reports OR escalated + rain $\geq 10\text{mm}$	Red pulsing marker
Cleared	Van collection event received	Green marker (reset)

Transitions:

Clear -> Reported: First citizen report for this dustbin

Reported -> Escalated: 3+ reports within 2hr window OR overflow ≥ 4

Escalated -> Critical: 5+ reports OR escalated state + rainfall $\geq 10\text{mm/hr}$

Any State -> Cleared: Admin marks van collection (overrides everything)

Cleared -> Clear: After collection, all reports are reset in the next window

8. API Reference

Method	Endpoint	Auth	Description
GET	/	-	Citizens' Portal (SPA)
GET	/admin	-	Admin Command Center
GET	/health	-	Production health check
GET	/api/config	-	Ward & dustbin registry
GET	/api/dashboard	-	Full cached Pathway state
GET	/api/dustbins	-	Dustbin registry + live status
GET	/api/forecast	-	3-day predictive risk forecast
POST	/api/report/dustbin/detect	-	Upload photo -> AI extraction
POST	/api/report/dustbin/confirm	-	Confirm detected ID
POST	/api/report/road-issue	Bearer	Admin: report road hazard
POST	/api/van/collection	Bearer	Admin: mark collected
POST	/api/van/clear-road	Bearer	Admin: clear road issue
POST	/api/demo/simulate-crisis	Bearer	Demo: inject crisis
WS	/ws	-	Real-time state broadcast

9. Event Lifecycle & Data Flow

The system processes events through a clear pipeline:

- **1. Photo Upload:** Citizen captures a photo of a waste issue via the portal
- **2. AI Detection:** FastAPI sends the image to Gemini 2.5 Flash Vision API
- **3. ID Extraction:** AI returns the exact MCD dustbin ID (e.g., MCD-W04-001)
- **4. Confirmation:** Citizen confirms the detection and submits the report
- **5. Event Write:** FastAPI writes a JSON event file to the data/reports/ directory
- **6. Weather Poll:** WeatherAPI.com rainfall data is polled every 10 minutes
- **7. Pathway Process:** Pathway detects the new file and triggers recomputation
- **8. Risk Scoring:** Risk scores are computed using the weighted multi-factor algorithm
- **9. State Machine:** Dustbin states are updated based on report counts and thresholds
- **10. Broadcast:** Atomic dashboard snapshot is broadcast via WebSocket to all clients
- **11. Admin Action:** Admin can dispatch vans, clear issues from the command center
- **12. Resolution:** Van collection events reset dustbin states and remove from queue

9.1 Rolling Windows

Waste reports use a 2-hour rolling window — events older than 2 hours are automatically expired. Road issues use a 6-hour window. This ensures the dashboard always reflects current conditions. The deduplication window is 5 minutes — same dustbin, same window = merged report.

10. Deployment Architecture

Component	Service	Tier
Web Server + Pathway Engine	Render.com Web Service	Free / Starter (\$7/mo)
AI Vision (Gemini 2.5 Flash)	Google AI Studio	Free tier (15 RPM)
Weather Data	WeatherAPI.com	Free tier (1M calls/mo)
CI/CD	GitHub Actions	Free (2000 min/mo)

Estimated Monthly Cost (Production): \$7-\$15/month for a single-city deployment.

10.1 Docker Deployment

The application is fully containerized with a Dockerfile. The docker-entrypoint.sh script starts both the Uvicorn ASGI server and the Pathway streaming engine as background processes. The container is deployed to Render.com via the render.yaml configuration file.

11. Security

Layer	Mechanism
Admin Endpoints	Bearer token authentication (strict 401)
Report Dedup	In-memory O(1) cache, 5-min window
Dustbin ID Validation	Strict regex MCD-Wd{2}-\d{3} against registry
Data Integrity	Atomic file writes (temp + rename)
CORS	Configurable origin whitelist

12. Scalability Path

Scale	Users	Architecture
Pilot (1 city)	10K	Single Render container (current)
Regional (10 cities)	100K	Horizontal Pathway workers + Redis pub/sub
National (100+ cities)	1M+	Kubernetes cluster, Kafka event bus

13. Project Structure

```
HACK-FOR-GREEN-BHARAT-HACKATHON/
+-- api/
|   +-- server.py           # FastAPI - transport only
+-- config/
|   +-- dustbins.py         # 72 MCD collection points
|   +-- wards.py            # 12 Delhi ward definitions
|   +-- settings.py         # Thresholds, windows, weights
+-- frontend/
|   +-- citizen.html/js/css  # Citizens' Portal (SPA)
|   +-- admin.html/js/css    # Admin Command Center
+-- llm_layer/              # Gemini integration module
+-- data/
|   +-- reports/            # Waste event JSON files
|   +-- road/               # Road issue event files
|   +-- vans/               # Van collection events
|   +-- weather/            # Weather polling data
|   +-- output/             # Pathway dashboard snapshots
+-- pathway_engine.py        # The Brain - all computation
+-- start.sh                # One-shot startup script
+-- requirements.txt         # Python dependencies
+-- Dockerfile              # Production container
+-- render.yaml             # Render.com deployment config
+-- .github/workflows/ci.yml # CI/CD pipeline
```

14. Setup & Installation

14.1 Requirements

- Python 3.10+ (Ubuntu WSL strongly recommended)
- Google Gemini API Key (<https://aistudio.google.com/> - free)
- WeatherAPI.com API Key (<https://www.weatherapi.com/> - free)

14.2 Installation Steps

```
git clone https://github.com/gintama1018/HACK-FOR-GREEN-BHARAT-HACKATHON.git
cd HACK-FOR-GREEN-BHARAT-HACKATHON
python3 -m venv .env
source .env/bin/activate
pip install -r requirements.txt
cp .env.example .env # Add your API keys
bash start.sh        # Start the server
```

Portal	URL
Citizens' Dashboard	http://localhost:8000/

Admin Command Center	http://localhost:8000/admin
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15. Demo Mode (For Judges)

The Admin Command Room includes a built-in 'Simulate Crisis' button. Pressing it injects 6 severe waste reports and a critical waterlogging road issue into Ward 12 (Shahdara North), triggering the full escalation matrix in real-time.

What the judges will see:

- Auto-triage the crisis into the Priority Queue
- Escalate dustbin states from Reported to Critical in seconds
- Render OSRM-routed road hazard polylines on the live map
- Apply weather multiplication if rainfall is detected
- Real-time WebSocket updates on both Citizens' and Admin portals simultaneously

16. Impact & Social Relevance

India loses over 3,500 lives annually to road accidents caused by potholes (MoRTH). The devastating floods in Punjab and Delhi exposed how open waste and blocked drainage amplify natural disasters into public health emergencies.

How InfraWatch Nexus addresses these crises:

- **Eliminating Reporting Friction:** A single photo replaces a 10-field government form. AI does the data entry. Citizens report in under 5 seconds.
- **Weather-Aware Prioritization:** A pothole during monsoon season is mathematically pushed to the top of the dispatch queue before it becomes fatal.
- **Optimizing Municipal Resources:** By clustering and deduplicating reports, city fleets target verified hotspots instead of patrolling blindly - reducing fuel waste and emissions.
- **Restoring Civic Trust:** Real-time map transparency proves to citizens that their government is responsive.

The goal is not to build another complaint box. The goal is to build a civic nervous system that feels danger before tragedy strikes.

17. Future Roadmap

Priority	Feature	Impact
P0	Negative sample awareness in AI prompts	Reduce false positives by 40%
P1	Spatial deduplication (Haversine)	Clean map, no cluster noise
P2	Auto-resolution of stale road issues	Self-healing map data
P3	7-category waste classification	Detailed analytics per ward
P4	Waste composition breakdown per ward	Policy-actionable insights
P5	Multi-city deployment (Kubernetes)	National scale platform
P6	Mobile app (Android APK)	Increased citizen reach

18. Research References

- Proenca & Simoes (2020). 'Deep Learning-Based Waste Detection in Natural and Urban Environments.' detect-waste benchmark, EfficientDet + EfficientNet two-stage framework.
- Mishra et al. (2025). 'iWatchRoad: Scalable Detection and Geospatial Visualization of Potholes for Smart Cities.' YOLOv8, OCR-GPS sync, Haversine dedup. arXiv:2508.10945.
- MDPI Applied Sciences (2024). 'IoT-Assisted Vehicle Route Optimization for Municipal Solid Waste Collection.' TOPSIS multi-criteria optimization, 14% route efficiency gain.
- IndiaAI / NITI Aayog (2024). 'AI-powered ward-wise performance reports for MSW management.' Ward-level analytics, predictive overflow with monsoon pattern integration.
- MCD Official Document: RO No. 20/DPI/MCD/2024-25. '106 Designated C&D Waste Collection Sites.' Municipal Corporation of Delhi (mcdonline.nic.in).
- Ministry of Road Transport & Highways (MoRTH). Road accident statistics and pothole-related fatalities data (2023-2024).